

Supplementary Material

AI-Assisted Tools in the Video Game Development Process: A Scoping Review (2020–2025)

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Appendix A. Coding Framework and Data Charting

Table S1: Coding scheme for Technical Aspect (TA)

Code	Category	Inclusion Criterion
T1	Machine Learning / Deep Learning	The study employs supervised, unsupervised, or deep-learning models (e.g., neural networks, clustering, regression) for automation, prediction, or classification tasks.
T2	Reinforcement Learning	The study implements reinforcement learning or dynamic policy optimization (e.g., Q-learning, deep-RL) applied to adaptive behavior or balancing.
T3	Natural Language Processing / Large Language Models	The work leverages NLP pipelines or LLMs (e.g., GPT, BERT) for dialogue generation, narrative assistance, or text understanding.
T4	Generative Models (GANs / Diffusion)	The system uses generative adversarial networks, diffusion models, or similar generative architectures to create new content or assets.
T5	Knowledge- or Rule-Based Systems	The approach relies on symbolic reasoning, expert systems, or rule-driven logic for decision making or content generation.
T6	Constraint- or Search-Based Methods	The method applies constraint solving, evolutionary search, or combinatorial optimization (e.g., CSP, GA, MOEA).
T7	Hybrid / Other AI Techniques	The study combines multiple AI paradigms or uses alternative approaches (e.g., heuristic systems, multimodal AI) not captured by T1–T6.

Table S2: Coding scheme for Methodological Design (MD)

Code	Category	Inclusion Criterion
M1	Experimental / Prototype Evaluation	The research tests a working system, algorithm, or prototype under controlled conditions or simulation.
M2	Case Study	The work documents the application of an AI method or tool in a specific real-world or production-like context.
M3	Survey / User Study	The paper gathers empirical data through surveys, usability testing, or user experience assessment.
M4	Conceptual / Framework Proposal	The study presents theoretical models, conceptual frameworks, or methodological proposals without direct experimentation.

Table S3: Coding scheme for Application Domain (AD)

Code	Category	Inclusion Criterion
D1	Game Development	The primary focus lies in video-game design, procedural content generation, gameplay balancing, or asset creation.
D2	Education and Training	The system or method is designed for learning, teaching, or serious-game applications supporting pedagogical outcomes.
D3	Software Engineering and Development Tools	The research targets software-engineering processes (e.g., testing, debugging, pipelines, or productivity aids).
D4	Art, Design, and Creative Media	The work addresses artistic creation, digital art, or cross-media design workflows beyond game-specific contexts.
D5	Other or Cross-Domain Applications	The study integrates AI techniques across mixed or emergent domains not covered by D1–D4.

Table S4: Coding scheme for Validation Approach (VA)

Code	Category	Inclusion Criterion
V1	User Testing / Usability Study	Validation includes participation of end users, designers, or players assessing usability, engagement, or satisfaction.
V2	Benchmarking / Quantitative Evaluation	Results are reported through quantitative metrics, benchmarks, or statistical comparisons.
V3	Expert Evaluation	Validation is conducted by domain experts, professional designers, or evaluators providing qualitative or semi-quantitative judgments.
V4	Comparative / Cross-Baseline Analysis	The study compares the proposed approach against baseline models, previous versions, or alternative methods.

Table S5: Coding scheme for Reported Contribution (RC)

Code	Category	Inclusion Criterion
C1	Framework / Architecture	The main output is a conceptual or software framework, design model, or system architecture.
C2	Tool / Prototype	The study delivers an implementable prototype, editor, or software tool evaluated in practice.
C3	Dataset / Benchmark Resource	The contribution includes data generation, open datasets, or benchmarking resources supporting replication.
C4	Review / Taxonomy / Mapping	The paper synthesizes prior research, offering a taxonomy, mapping study, or structured literature review.

Table S6: Coding summary of included studies 2020–2024 ($n = 67$).

No.	Short Title (GitHub filename)	TA	MD	AD	VA	RC
1	PCG for Digital Twins (3.pdf)	T1, T6	M1	D1	V2	C2
2	AI-Powered Game Design with Chat-GPT (13.pdf)	T3	M1	D1	V1, V3	C2
3	Surrogate Model for FPS PCG (15.pdf)	T1, T4	M1	D1	V2	C2
4	Procedural Puzzle Generation (23.pdf)	T1	M1	D1	V1	C2
5	Human-AI Co-Creation Framework (24.pdf)	T7	M4	D5	V3	C1
6	Affective 3D Scene Generation (26.pdf)	T3, T4	M1	D4	V3	C2
7	Baba is You Mixed-Initiative Tool (27.pdf)	T1, T6	M1	D1	V1	C2
8	RL-Based Multiplayer Balancing (28.pdf)	T2, T1	M1	D1	V2	C2
9	AI-Assisted PCG Tools (29.pdf)	T1	M1	D1	V1, V3	C2
10	Constraint-Based Tile Map Generator (30.pdf)	T6	M1	D1	V2	C2
11	Hybrid Search PCG (34.pdf)	T7	M1	D1	V2	C2
12	FPS Level Generation with GANs (36.pdf)	T4	M1	D1	V2	C2
13	Mixed-Reality Asset Assistant (38.pdf)	T1, T7	M2	D4	V1, V3	C2
14	Game Logic Synthesis via ML (39.pdf)	T1	M1	D3	V2	C2
15	Ontology-Based Game Content Generation (40.pdf)	T5	M1	D1	V2	C1
16	Player Modeling via Logs (41.pdf)	T1	M1	D1	V2	C2
17	Narrative Quality Estimation (42.pdf)	T3	M2	D1	V3	C2
18	AI-Driven Narrative Generation (43.pdf)	T3	M2	D1	V3	C2
19	Constraint-Based Dungeon Generator (44.pdf)	T6	M1	D1	V2	C2
20	ChatGPT for Designer Workflows (45.pdf)	T3	M3	D1	V1, V3	C2
21	Digital Twin PCG Workflow (46.pdf)	T7	M2	D5	V2	C2
22	Player Experience Prediction (47.pdf)	T1	M1	D1	V2	C2
23	Evolutionary Level Variety (48.pdf)	T6	M1	D1	V2	C2
24	Narrative Branching Helper (49.pdf)	T3	M3	D1	V1, V3	C2
25	Asset Tagging with Computer Vision (50.pdf)	T1	M1	D3	V2	C2
26	LLM Dialogue Drafting (51.pdf)	T3	M3	D1	V1	C2
27	Visual Style Transfer for Assets (52.pdf)	T4	M1	D4	V2	C2
28	Automated Bug Reproduction (53.pdf)	T1	M2	D3	V2	C2
29	Human-AI Co-Design Patterns (54.pdf)	T7	M4	D1	V3	C1
30	Knowledge-Based Quest Generator (55.pdf)	T5	M1	D1	V2	C1
31	Player Experience Estimation (56.pdf)	T1	M3	D1	V1, V2	C2
32	Procedural City Layouts (57.pdf)	T6	M1	D1	V3	C2
33	Texture Synthesis for Game Assets (58.pdf)	T4	M1	D4	V2	C2
34	Prompted Narrative Beats (59.pdf)	T3	M3	D1	V1	C2
35	Pathfinding Data Generator (60.pdf)	T1	M1	D3	V2	C3
36	Telemetry-Based Balancing (61.pdf)	T1	M1	D1	V2	C2
37	Game QA Prioritization (62.pdf)	T1	M2	D3	V2	C2
38	Puzzle PCG with Constraints (63.pdf)	T6	M1	D1	V2	C2
39	Designer Feedback PCG (64.pdf)	T6	M1	D1	V2	C2
40	NPC Behavior Synthesis (65.pdf)	T1	M1	D1	V1	C2
41	Impressions of the GDMC AI Settlement Generation Challenge in Minecraft (66.pdf)	T7	M3	D1	V3	C1
42	Adaptive Difficulty via RL (70.pdf)	T2	M1	D1	V2	C2

No.	Short Title (GitHub filename)	TA	MD	AD	VA	RC
43	RL Dynamic Difficulty Adjustment (71.pdf)	T2	M1	D1	V2	C2
44	Multi-Objective EA for Levels (77.pdf)	T6	M1	D1	V2	C2
45	Evolutionary Level Optimization (79.pdf)	T6	M1	D1	V2	C2
46	Model-Based NPC Testing (91.pdf)	T1	M2	D3	V2	C2
47	LLM QA Assistant for Games (92.pdf)	T3	M2	D3	V1, V3	C2
48	Procedural Terrain GAN (107.pdf)	T4	M1	D1	V2	C2
49	Playability Predictor (108.pdf)	T1	M1	D1	V2	C2
50	PCG Evaluation Toolkit (111.pdf)	T6	M1	D1	V2	C3
51	Procedural Texture GAN (112.pdf)	T4	M1	D4	V2	C2
52	Knowledge-Graph Quest Design (122.pdf)	T5	M1	D1	V2	C1
53	Interactive PCG Editor (126.pdf)	T7	M1	D1	V1, V2	C2
54	PCG Diversity Controller (132.pdf)	T1, T6	M1	D1	V2	C2
55	Hybrid Narrative Planner (135.pdf)	T5, T3	M1	D1	V3	C1
56	Novelty Search for PCG (137.pdf)	T7	M1	D1	V2	C2
57	Sketch-to-Level Co-Creation (138.pdf)	T1, T7	M3	D1	V1	C2
58	AI for Educational Game Design (143.pdf)	T1, T3	M1	D2	V1	C2
59	LLM-Based Narrative Game Design (144.pdf)	T3	M3	D2	V1, V3	C2
60	GPT Dialogue Tool for Indie Games (145.pdf)	T3	M1	D1	V1, V3	C2
61	Narrative Assistants for Game Design (146.pdf)	T3	M3	D1	V1, V3	C2
62	SPHINX Puzzle Generator (147.pdf)	T1, T6	M1	D1	V2	C2
63	Affective Modeling for Game Design (148.pdf)	T3	M1	D1	V3	C1
64	Generative Visual Art for Games (150.pdf)	T4	M1	D4	V3	C2
65	Diffusion-Based Mario PCG (151.pdf)	T4	M1	D1	V2	C2
66	V4Design Creative Toolkit (152.pdf)	T7	M4	D4, D5	V2	C2
67	AI Dialogue Systems for Games (158.pdf)	T3	M1	D1	V1	C2

Table S7: Coding summary of included studies for 2025 (n = 68).

No.	Short Title (GitHub filename)	TA	MD	AD	VA	RC
1	Progression Balancing x Baldur's Gate 3 (3706598.3713162.pdf)	T2, T7	M1, M3	D1	V1, V2	C1, C2
2	Support Autonomy: Exploring Player Perspectives on AI-Supported Onboarding (3706598.3713576.pdf)	T5	M3	D1	V1	C2
3	Archaeological Gameworld Affordances (3706598.3714036.pdf)	T6	M3, M4	D1	V1	C1, C2
4	Dreamcrafter: Immersive Editing of 3D Radiance Fields (3706598.3714312.pdf)	T4	M1, M3	D4	V1, V2	C2
5	Adaptive Difficulty & Personalized Player Experience (3743049.3743070.pdf)	T1	M1	D1	V2	C1
6	GRAB: Griefing Recognition via Anomaly Behavior (3744736.3749354.pdf)	T1	M1	D3	V2	C2
7	DRL for Optimizing Game Strategies (3745812.3745844.pdf)	T2	M1	D1	V2	C1
8	AI-Driven NPCs for Player Challenges (3749012.3749054.pdf)	T1	M1	D1	V1	C2
9	IDN Authorial Burden Framework (3757746.pdf)	T3	M3, M4	D1	V3	C1
10	GenAI for Emotional Engagement in Games (3770445.3770463.pdf)	T3, T4	M1	D1	V1	C2

No.	Short Title	TA	MD	AD	VA	RC
11	Automatic Identification of Game Stuttering via Gameplay Videos (.3695992.pdf)	T1, T7	M1	D3	V2	C2
12	A Comprehensive Model of Automated Evaluation of Difficulty in Platformer Games (.3705013.pdf)	T5, T7	M1	D1	V1, V2	C1, C2
13	DreamGarden: A Designer Assistant for Growing Games from a Prompt (.3706598.3714233.pdf)	T3, T4	M1, M3	D1	V1, V3	C2
14	Enhancing UX Evaluation of Graphic Art Style Games with GenAI (.3708036.3708042.pdf)	T4	M3	D1, D4	V1, V3	C2
15	The Many Challenges of Human-Like Agents in Virtual Game Environments (.3709347.3743837.pdf)	T3, T7	M4	D1	V4	C4
16	Malinowski in the Age of AI: LLMs for Text Game Creation (.3719236.3719240.pdf)	T3	M1	D1	V3	C2
17	Celeste AI Framework: Automated 2D Level Generation (.3723498.3723796.pdf)	T1, T6	M1	D1	V2	C2
18	Slice of Life: Symbolically Grounded LLM Generative Dialogue (.3723498.3723806.pdf)	T3, T5	M1	D1	V1	C2
19	Constraint Is All You Need: 3D Level Generation with LLMs (.3723498.3723840.pdf)	T3, T6	M1	D1	V2	C2
20	Democratizing Game Modding with GenAI (.3748612.pdf)	T3, T4	M2, M3	D1	V1	C2
21	Mixed-Initiative Co-Creativity Tutorial (.3759914.pdf)	T7	M4	D1	V4	C4
22	Remaking Game Art Based on Deep Learning (.3772726.3772850.pdf)	T1, T4	M1	D4	V2	C2
23	Use of AI in Game Development (Review) (.3773365.3773548.pdf)	T7	M4	D1	V4	C4
24	Game Intelligence AI Fusion Framework (.3777730.3777806.pdf)	T7	M4	D1	V4	C1
25	GenAI in Game Development Bibliometric Study (.3777730.3777843.pdf)	T4, T7	M4	D1	V4	C4
26	Systematic Review of GenAI on Character Creation (A.Systematic.Review...pdf)	T3, T4	M4	D1, D4	V4	C4
27	A Generative AI Game Jam Case Study (313.pdf)	T4	M1	D1, D4	V2, V4	C2
28	3D Facial Speech Animation Gen for Games (.162.pdf)	T4	M1	D4	V2	C2
29	Hybrid Co-Finetuning for Visual Bug Detection (.169.pdf)	T5	M1	D3	V2	C2
30	Visualizing Possibility Spaces of DDA Systems (.170.pdf)	T1, T7	M1, M2	D1, D3	V3, V4	C2
31	Program Synthesis for Interpretable Game AI (.171.pdf)	T2	M1, M3	D1	V4	C2
32	Enhancing Game AI Behaviors with LLMs (.172.pdf)	T2, T3	M1	D1	V2	C2
33	GenAI for Graphic Art Style UX Evaluation (.173.pdf)	T4	M1	D4	V2, V4	C2
34	AI-Driven NPC Dialogues in Unity (.191.pdf)	T3	M1	D1	V1, V2	C2
35	Procedural Gen of Race Track Surroundings (.197.pdf)	T6	M1	D1	V2	C2
36	Diversity-Oriented Testing for Game Agents (.198.pdf)	T2, T5	M1	D3	V2	C2
37	Goal-Oriented Interactions in Games Using LLMs (.208.pdf)	T3	M1	D1	V2	C2
38	NPC Behavior in Games Using Unity ML-Agents (.223.pdf)	T2	M1	D1	V2	C2

No.	Short Title	TA	MD	AD	VA	RC
39	Procedural Game Content Generation using ML (.228.pdf)	T4, T6	M1	D1	V2	C2
40	Radial-Layered Procedural Dungeon Gen (.231.pdf)	T6	M1	D1	V2	C2
41	Procedural level generation with GANs Review (.236.pdf)	T4, T6	M4	D1	V4	C4
42	Enhancing character animation with GANs (.241.pdf)	T4	M1	D4	V2	C2
43	ScriptDoctor: Auto Gen of Puzzle-Script (.242.pdf)	T3, T5, T6	M1, M3	D1, D3	V1, V2	C2
44	Enhancing Game AI Reasoning with LLMs (.248.pdf)	T1, T3	M1	D1	V1, V2	C2
45	Using Planning for Automated Game Testing (.326.pdf)	T5	M1	D3	V2	C2
46	Emotionalyzer: ML for Video Game Testing QA (.332.pdf)	T1, T7	M1	D3	V2	C2
47	Group behavior simulation for narrative generation (.334.pdf)	T1, T3	M1	D1	V1, V2	C1, C2
48	Challenges of Human-Like Agents in Games (.166.pdf)	T2	M4	D1	V4	C4
49	GNNetic: Player Decisions to Drive Interface Design (.181.pdf)	T1	M1	D1, D3	V2	C2
50	The Use of AI Technology in Game Development (.184.pdf)	T7	M4	D1, D3	V4	C4
51	AntiCheatPT: Cheat Detection in Competitive Games (280.pdf)	T1	M1	D3	V2	C2, C3
52	AutoGT: Automatic Generation of Game Trees (281.pdf)	T6	M1	D1	V2	C2, C3
53	Analyzing Enemy Behavior Difficulty (.279.pdf)	T5	M1	D1	V2	C1, C2
54	Contextual Tutorial Generation via MCTS (.296.pdf)	T1, T6	M3	D1	V1, V2	C2
55	Visual Perception System for Game Agents (.304.pdf)	T7	M1	D1	V2	C2
56	Survey on Human-Centered Agents for Fighting Games (.278.pdf)	T1, T2	M4	D1	V4	C4
57	Generative AI in PCG for Games (Review) (.286.pdf)	T4, T6	M4	D1	V4	C4
58	Human-Like Bots for Tactical Shooters (.287.pdf)	T1	M1	D1	V2	C2
59	DL for Image-Based Game Asset Gen (Review) (.288.pdf)	T1, T4	M4	D4	V4	C4
60	LLM Auto Commentary Generation in Games (.290.pdf)	T1, T3	M1	D1	V2	C1, C2
61	WFC for Puzzle-Level Generation (.294.pdf)	T6	M1	D1	V2, V4	C2
62	Player Tactics via Sequence Mining (.298.pdf)	T1	M2	D1	V3	C1
63	DRL for Dynamic NPC Behavior in Unity (.299.pdf)	T2	M1	D1	V2	C2
64	Verbal Harassment Detection in MMOs (.300.pdf)	T1, T3	M1	D3	V2	C2
65	Dynamic Difficulty Adjustment SLR (.301.pdf)	T7	M4	D1	V4	C4
66	AI Artificial Experience in Playtesting (.305.pdf)	T3	M1	D1, D3	V4	C2
67	Real-Time Player Engagement Telemetry (.311.pdf)	T1	M1	D1	V2	C2
68	Across-Game Engagement via Few-Shot Learning (.330.pdf)	T1	M1	D1	V2, V4	C1

Appendix B: Full List of Excluded Articles

Table S8: Full list of excluded studies after full-text screening for the period 2020–2024

#	Short Title	Exclusion Code
1	DALL-E Mini and Surrealism	E2
2	AI Bilingual Education	E2
3	FSM Adaptive Learning	E2
4	Declarative PCG for TTRPGs	E1, E2, E3
5	Mixed-Initiative Chart Editor	E3
6	Collaborative VR Training Game	E2
7	Educational Maze Game Designer	E3
8	AI-Assisted Recruiting Tools	E2
9	AI Highlights for Sports Media	E2
10	Board Game-Based Learning with AI	E2
11	Natural Language Dialogue with Robot NAO	E2, E6
12	Story & Music Generation Engine	E3
13	Children’s Emotion Learning with AI	E2
14	Structural Damage ID via ELM	E2
15	Procedural Geodesign for Ecosystem Services	E2
16	Peer Learning in IT Education	E2
17	Detecting AI Cheating in Go	E2
18	Algebraic NEAT Program Design	E2
19	AI for Melanoma Diagnosis	E2
20	Computer Aided Diagnosis of Melanoma	E2
21	Gaming for Parkinson Assessment via Movement Analysis	E2
22	AI in Oncology Imaging Through Game-Theoretic Models	E2
23	AI and Law	E2
24	Children’s Emotion Learning with AI	E2
25	Game Theory in Negotiation Platforms	E2
26	Narrative Visualization Tools	E2
27	Training Efficiency in Casinos	E2
28	Social Games for Intercultural Training	E2
29	Language Learning Game Study	E2
30	AI in Music for Cognitive Tasks	E2
31	Gamified Streaming Labor	E2
32	Decentralized Futures in Gaming	E2
33	Governance in VR Spaces	E2
34	Ethical Frameworks for Game Design	E2
35	AI Smart Ads in Games	E2
36	AI Feedback in Fitness Games	E2
37	AI for Sports Commentary	E2
38	Mixed-Initiative Music Composition	E1, E4
39	HarmonyMapper Emotion Model	E1, E3
40	AI-Assisted Music Classification	E7
41	LSTM PCG for Minecraft	E1
42	LLM-based Level Design from Descriptions	E1, E3
43	Voice-Based Interactive Story	E1, E2
44	Evaluating Generative Tools with Players	E4
45	Fairness-Aware NPC Behavior	E2, E3
46	Story Evolution System for VR	E1, E3
47	AI for Learning in Games with Dyslexia	E2
48	LLM-Based Plan Generation in Learning Games	E2
49	Digital Culture in NPC Design	E2
50	ChatGPT for Creative Writing in Games	E3
51	LLMs in Gamified Learning Platforms	E2
52	AI for Emotion Tracking in Education	E2
53	Nonscalability in No Man’s Sky	E2
S 54	Economic Analysis of GenAI Platforms	E2
55	Game Culture in Chinese GenAI Ecosystems	E3
56	StoryBrush: Text-to-Image for Comics	E1, E2
57	Legal Implications of Generative AI	E2, E3
58	AI and Emotions in HRI Games	E2
59	Ethics of Virtual Creatures	E2
60	LLMs for Therapeutic Gaming	E2
61	ChatGPT in Science Education Games	E2
62	LLM-Based Serious Games for Democracy	E2
63	Designing Diverse NPC Behaviors in Open Worlds	E3

#	Short Title	Exclusion Code
64	Explainable AI in Game Dialogue	E3, E4
65	SimpleG: A Beginner-Friendly PCG Tool	E2
66	Bias in Game Datasets for AI Agents	E2, E4
67	Using ChatGPT to Enhance Player Immersion	E2, E3
68	NarSUM: AI Summarization in eSports	E3, E4
69	Affect Modeling with Transformers in Games	E2
70	FSM Traces for Robot Behaviors	E2
71	Rule Induction for Strategic Game Play	E2, E3
72	AI Feedback in Educational Game Platforms	E2
73	Text-to-Movie Authoring of Anatomy Lessons	E2
74	AI-Driven Story Understanding in Films	E2
75	Ethics in LLM-based Game NPCs	E3
76	Procedural Textures with GANs for Education	E2
77	Creative Coding and AI Storytelling	E2
78	Emotive AI for Digital Art	E2
79	Playful AI in Education Games	E2
80	Evaluating AI Chat for Game Tutoring	E2
81	Reinforcement Learning in Training Sims	E2
82	LLMs for Emotional Companion Characters	E3
83	Cultural Expression through Gameplay Logs	E3

Table S9: Full list of excluded studies after full-text screening (2025 update)

#	Short Title	Exclusion Code
84	Cartoonimator: Tangible Kit for Keyframe Animation	E2
85	WebTraceSense for User Interaction Log Analysis	E2
86	KINAITICS: Enhancing Cybersecurity Education	E2
87	Autistic Adults in Participatory Design	E1, E2
88	Escape or D13: Youth Perspectives of AI	E1, E2
89	LogoMotion: Visually-Grounded Code Synthesis	E2
90	Conversational GenAI for Youth Mental Health	E2
91	Enhancing AI Explainability with Gamification	E2
92	Generative AI in VRChat Communities	E1, E3
93	Scaling Analysis via Fuzzy Linkography	E1, E3
94	SLIMO CITY: Bio-HCI Game System	E2
95	Equality Engine: ML Bias Literacy Game	E2
96	FairPlay: Collaborative Bias Mitigation in Datasets	E2
97	Digital Transformation in Higher Education	E2
98	STEM Digital Textbook Based on Metaverse and GenAI	E2
99	Quality Diversity GP for Scheduling Heuristics	E2
100	Unleashing Creativity in the Metaverse: GenAI	E1, E2
101	Collaborative Co-Creation with AI in Music Production	E2
102	Serious Games for Computing Education	E2
103	Literature Review of Conversational User Interfaces	E2
104	Digital Tools for Cultural Heritage Preservation	E2
105	Multi-agent Multi-game Entity Transformer in MARL	E2
106	Sensemaking Play of Jubensha Games	E1
107	AIGC Technology in Digital Media Companies	E2
108	Dice Adventure: Hybrid Teaming Effects	E1, E2
109	Computational Tools for Table-Top RPGs: Scoping Review	E2
110	Quack the Code: Learning Through Teaching AI	E2
111	AIGC for Home Textile Designers	E2
112	TrackID3x3: Tracking in Basketball	E2
113	Trust Region-Based Bayesian Optimisation	E3
114	Future of Games in Society Workshop	E1, E3
115	E-Participation Urban Planning	E2
116	Mixed Reality for 3D Design	E2
117	Gamification for Employees with ADHD	E2
118	POET: Text-to-Image Generation Prompting	E2
119	GNNetic: Player Decisions in Card Games	E2
120	Generative AI and TTRPG Hobbyists	E2
121	Unveiling Ruby: Stack Overflow Insights	E2
122	AI-Driven Systematic Reviews	E2
123	Meaning-Typed Language Abstraction	E2
124	Gamified Crowd-sourcing for Disambiguation	E2
125	Programming Education Research Analysis	E2

#	Short Title	Exclusion Code
126	Player Dynamics in Battle Royale	E1
127	Machine Learning Systems Survey	E2
128	Explainable AI Frameworks Review	E2
129	Autism Computing Research	E2
130	Scientometric Analysis of CHI	E2
131	GAS Workshop Insights	E3
132	Decoding the Chameleon Game	E2
133	BERT for Apples to Apples	E2
134	Biases in Human Wordle Gameplay	E2
135	Cognitive Load with Wearable Sensors	E2
136	Spread of Game Innovations (Review)	E1, E2
137	RAG for Regulatory Compliance	E2
138	Scientific Visualizations Uncertainty	E2
139	Serious game for hospitalized children	E2
140	ML Applications in MOBA Esports	E2
141	Facial emotion detection & classification	E2
142	Video-Enhanced Offline RL	E2
143	MCU: Evaluation Framework for Game Agents	E2
144	Games for Computational Concepts Framework	E2
145	Nordic Perspectives on Algorithmic Systems	E1, E2
146	Gamified Learning Method for AI Programming	E2
147	Digital Twins and 3D Simulations with UE5	E3
148	Online Gaming on Occupational Recovery	E2
149	ML to Predict Effects of Games on Youth Behavior	E2
150	AR Indoor Navigation for Cognitive Inclusion	E2
151	Industrial Applications of Q-Learning	E2
152	Retroductive Perspective on Games	E1, E3
153	EEG Signal for Game Video Emotion	E2
154	Automated UI Generation via Diffusion	E2
155	Digital games into teaching	E2
156	VR field trips	E2
157	Gamification in Engineering Education	E2
158	AI-generated videos in medical education	E2
159	BCI for EEG Motor Imagery	E2
160	Machine Learning in MILAGE LEARN+	E2
161	LLMs for Educational AI Interactions	E2
162	Gamification in Financial Services	E2
163	GTA-HDR Dataset for Image Reconstruction	E2
164	Hybrid Autonomous Navigation (RRT & RL)	E2
165	Video Game Sales Prediction with ML	E2
166	Intervention platform for Type 1 Diabetes	E2
167	Visual representation of health care data	E2
168	Discriminating Experts Using Blinks	E2
169	Mobile Game UI Usability Evaluation	E1
170	Brain stimulation for gaming disorder	E2
171	Adaptive AI in elderly healthcare	E2
172	Real Time Gameplay using Pose Detection	E1, E2
173	Engagement levels in gamified environments	E2
174	PUBG gaming, sleep disorder, and migraine	E2
175	Myoelectric pattern recognition for stroke	E2
176	Gamified binocular therapy	E2
177	GenAI and gamification for biomedical treatment	E2
178	LLM-Driven Adjustments in Serious Games	E2
179	VR with Moodle for Telemedicine	E2
180	LLM in Serious Games for Anorexia Caregivers	E2
181	Markerless Telerehabilitation	E2
182	Code Refactoring in Roblox	E1
183	Prompt-Driven Dynamics in Serious Games	E2, E3
184	Serious Games for Computational Thinking	E2
185	Visual Explanation in DRL with AR	E2
186	Cross-domain recommendations with PMF	E2
187	Virtual escape room for statistics learning	E2
188	Grain boundary network optimization	E2
189	3D model classification with MetaFormer	E2
190	Simulation of street landscape design	E2
191	Commercial games to teach coding	E2
192	Societal impact of AI in game software	E1, E3

#	Short Title	Exclusion Code
193	How players develop competitive skills	E1
194	NLP to identify ecogames for education	E2
195	Gamification in a college course	E2
196	RL algorithms in FPS using VizDoom	E2
197	What is a game review: Steam analysis	E2
198	TIMECRAX History Game	E2
199	Waste Separation Learning Game	E2
200	Advancing Video Game Addiction Detection	E2
201	Analysis of the Student Experience	E2
202	AntiCheatPT: Transformer Cheat Detection	E2
203	Artificial Intelligence as a Director	E3
204	CSG-based ML 3D Sketch Translation	E2
205	Comparison of Gamified Binocular Therapy	E2
206	Brain-Computer Interfaces in Gaming	E2
207	Gamification Frameworks for Behavioural Change	E2
208	Immersive Games for C-STEAM Programs	E2
209	Game Design Engineering Process Centered on PX	E1
210	VR Game-Based Classification of Arachnophobia	E2
211	Game Prototype with GIMs for Neurodiversity	E2
212	NBA 2K20 Player Rating Predictions	E2
213	Interactive Treasure Hunts via LLM	E2
214	AI Director in Horror Games (Player Survey)	E1, E3
215	GenAI-Driven Innovation in Game Dev	E3
216	Equality Engine: Fostering ML Bias Literacy	E2
217	COVID-19 Fighter: Interactive Learning Platform	E2
218	Dashboard Design in Serious VR Game for Cognitive Rehab	E2
219	Design Aesthetics in Post-Digital Age (Turkish)	E6
220	Design Thinking for Game Design Course (Teaching Paradigm)	E2
221	Digital Game to Assess Students' Information Literacy	E2
222	Educational Video Games and Meaningful Learning in Mathematics	E2
223	EEG Risk Markers of Gaming Addiction (ML + Iowa Gambling Task)	E2
224	Enhancing Gamified Training Usability for Professional Learning	E2
225	Flipped Classroom and Multimedia to Motivate Programming	E2
226	Game-Based Learning Analytics (Integrated Design Process)	E2
227	Game Design Principles from Players' Negative Feedback	E1
228	Gamification for Talent Management in Industry 5.0	E2
229	Gamification in Tourism Research (Systematic Review)	E2
230	Gamification within Higher Education	E2
231	Gamified Learning for Cybersecurity Skills	E2
232	Gamifying MITRE ATTACK for Cybersecurity Training	E2
233	Gaming in Higher Education (Distributed Leadership)	E2
234	Human-Robot-Game Platform for Autism (Facial Expression Analysis)	E2
235	Mesh Smoothing Techniques	E2
236	ML Model for Myopia Factors in Students	E6
237	Motivational Qualities within Serious Game Development	E3
238	Myoelectric Pattern Recognition + VR for Stroke Rehab	E2
239	Online Gaming and Mental Distraction (RNN vs Lasso)	E2
240	Online Gaming to Avoid Mental Distraction (RNN vs ANN)	E2
241	Playability and Player Experience in Smartphone Games	E1
242	Predict Effects of Gaming on Growth Mindset (Steam Reviews)	E1
243	Preservice Teachers with AR/AI Hands-on Learning	E2
244	Product Interaction Ratings via Emotion Detection	E2
245	Product Placements on Digital Platforms (Marketing)	E2
246	Serious Games and Assistive Robotics for Neurodevelopmental Disorders	E6
247	Skill Development and Assessment through Game-Based Learning (Review)	E2
248	Teachers Using GenAI Tools in Gamified Learning Activities	E2
249	Water Culture via AR and Gamified Experiences	E2
250	AR Games in Autonomous Driving	E2
251	PandaSkill - Esports Skill Rating	E2
252	XR Biofeedback Training	E2
253	ML for Adolescent Depressive Symptoms	E2
254	AI Video-games for Stroke Dysphagia	E2
255	Clinical Decision Support via Game Biomarkers	E2
256	Preschool Learning with AI Toys	E2

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257	Gamified Speech Training for Students	E2